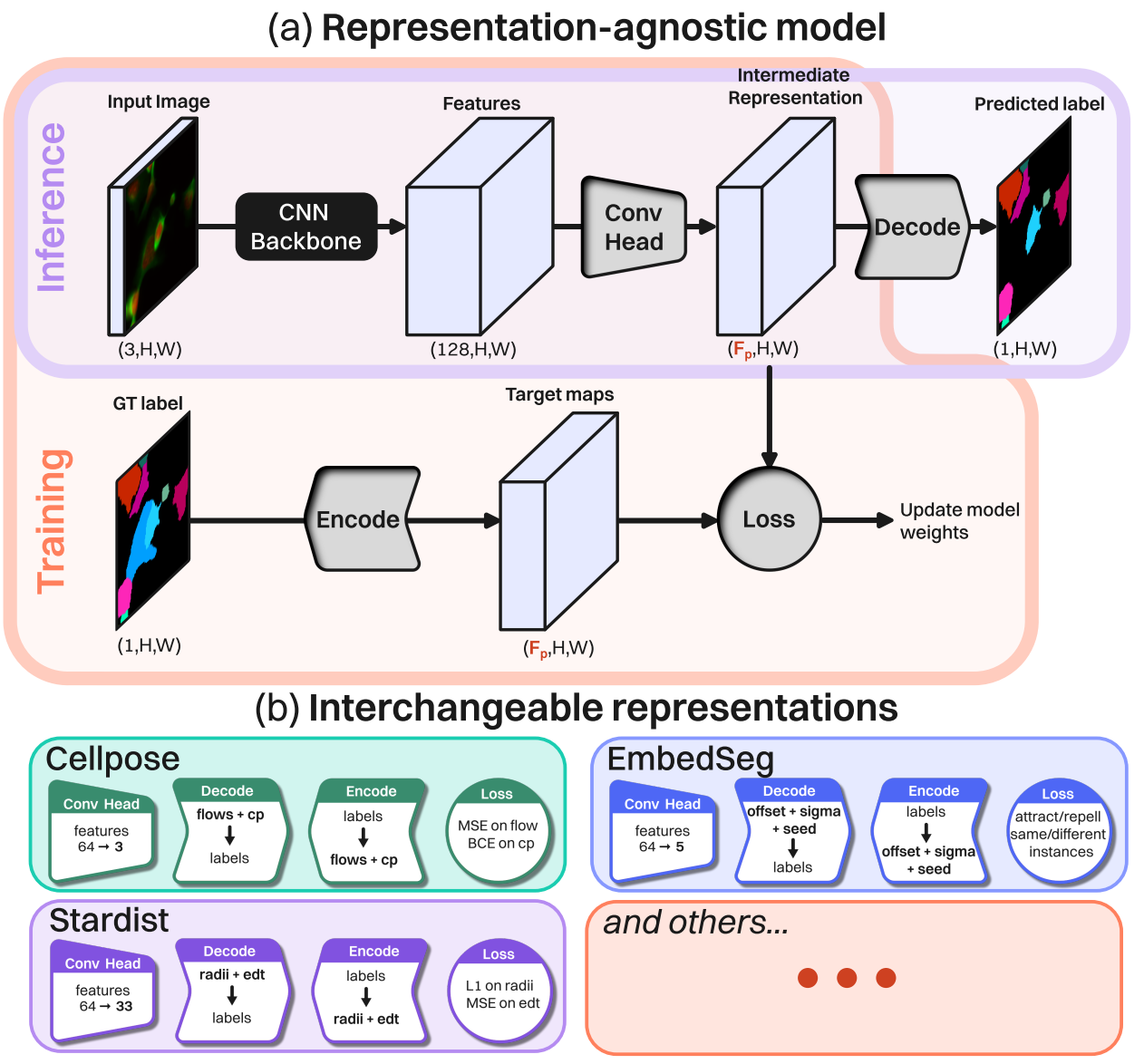

(a) Representation-agnostic model
Inference
Input Image
CNN Backbone
Features
Conv Head
Intermediate Representation
Decode
Predicted label
(3,H,W)
(128,H,W)
(Fp,H,W)
(1,H,W)
Training
GT label
Encode
Target maps
Loss
Update model weights
(1,H,W)
(Fp,H,W)
(b) Interchangeable representations
Cellpose
Conv Head
features 64 → 3
Decode
flows + cp
labels
Encode
labels
flows + cp
Loss
MSE on flow
BCE on cp
EmbedSeg
Conv Head
features 64 → 5
Decode
offset + sigma + seed
labels
Encode
labels
offset + sigma + seed
Loss
attract/repell same/different instances
Stardist
Conv Head
features 64 → 33
Decode
radii + edt
labels
Encode
labels
radii + edt
Loss
L1 on radii
MSE on edt
and others...